GLS versus Averaged Change-Score Estimators for Clinical Trials with Run-In Observations

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Abstract

Background. Three estimators of the treatment effect in longitudinal trials with run-in observations are in routine use: the GLS estimator under the linear mixed model, which exploits the full covariance structure to extract maximum information from all observations; the ANCOVA estimator, which regresses post-randomization means on the run-in mean as a covariate without specifying the random-effects covariance; and the averaged change-score estimator, which computes a simple difference of phase means and applies a t-test, requiring no variance-component estimation. The relative efficiency of these estimators under correct and misspecified models has not been comprehensively characterized.

Methods. We derive the variance of the three estimators as a function of the trial design (J_0, J_1, J_2) and the variance parameters (σ^2, σ_b^2). The derivation proceeds via the Woodbury-based variance formulas of Paper 1 in this compendium, plus closed-form expressions for the ANCOVA and averaged-change-score variances under the same random-slopes model. Robustness to misspecification of the within-subject covariance (R) and the random-effects covariance (G) is assessed analytically through sandwich variances, and a Monte Carlo simulation under compound symmetry evaluates the estimators alongside a GLS analysis that misstates the random-slope variance.

Results. All comparisons are made on the common estimand scale, with the averaged and ANCOVA contrasts rescaled to unbiasedness for the treatment effect γ. Under correct specification, GLS $\leq$ ANCOVA $\leq$ averaged in variance, with no violations over a 115-cell grid of designs and signal-to-noise ratios. The efficiency penalty of the simpler estimators is substantial in run-in designs with multiple treatment-period visits: at $(J_0, J_1, J_2) = (2, 2, 0)$ the relative efficiency of ANCOVA versus GLS is 0.64 to 0.81 across signal-to-noise ratios, an efficiency loss of 19% to 36%. Relative efficiency above 0.95 occurs only in single-post-visit run-in designs ($J_1 = 1$, $J_2 = 0$) and in designs without run-in or common close ($J_0 = 0$, $J_2 = 0$) at high signal-to-noise. In the MIRIAD-based example the ANCOVA sample size exceeds the GLS sample size by up to a factor of roughly 2.6 among run-in designs, and the averaged sample size by up to a factor of roughly 45. Under the G and R misspecifications studied, GLS never loses its advantage over the simpler estimators: the misspecified-GLS sandwich variance peaks at 5.3 against an averaged-estimator variance of 8.9 at the reference design.

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Conclusions. GLS is the efficient estimator under correct specification, and its advantage survives the covariance misspecifications examined here. The averaged change-score estimator remains useful as a fully prespecified robust benchmark and a teaching tool. The open robustness question, not studied here, is whether feasible GLS with variance components estimated from small samples retains this advantage.

Contents

1 Introduction

[1] **The optimality of GLS is contingent on knowing the covariance, which in practice must be estimated and may be unstable in small samples.**

- GLS is the best linear unbiased estimator (BLUE) only when the covariance matrix is known.
- Feasible GLS substitutes estimated variance components, so its efficiency depends on estimation accuracy.
- Small samples or short follow-up yield unstable estimates that can erode the theoretical advantage of GLS.

The GLS estimator $\hat{\gamma}_{GLS} = (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}\mathbf{c}$ is the best linear unbiased estimator (minimum variance among linear unbiased estimators) when Σ is known. In practice, Σ must be estimated, and the efficiency of the feasible GLS depends on the accuracy of this estimation. With small sample sizes or short follow-up, variance component estimates can be unstable, potentially eroding the theoretical advantage of GLS.

The averaged change-score approach sidesteps variance component estimation entirely. For a trial with J_0 run-in and J_2 common close observations, the estimator is:

$$\hat{\gamma}_{\text{avg}} = \bar{d}_1 - \bar{d}_0 \tag{1}$$

where

$$d_{ik} = \frac{1}{J_1 + J_2} \sum_{\ell=1}^{J_1+J_2} Y_{i,\ell,k} - \frac{1}{J_0} \sum_{\ell=1}^{J_0} Y_{i,-\ell,k} \tag{2}$$

is the difference of the post-randomization and run-in means for participant i. The baseline visit is the change-score reference and cancels, so it enters neither average; this matches the contrast implemented in `var_gamma_avg()`. Throughout, the code and figure parameters use (r, k, f) for the design counts, with the correspondence $J_0 = r$ (run-in), $J_1 = k$ (treatment period), and $J_2 = f$ (common close); the prose and table headers use (J_0, J_1, J_2).

[2] **The relative merits of change-score and ANCOVA analyses have been studied extensively across several methodological traditions.**

- Formal efficiency comparisons for pretest-posttest designs trace back to @brogan1980 and @yang2001.
- @frison1992 and @frison1997 established the efficiency ordering among post-treatment, change-score, and ANCOVA analyses with multiple baseline measurements, the closest antecedent of the present comparison.
- Subsequent work contrasted baseline-as-covariate with baseline-as-response and developed semiparametric covariate adjustment.
- Later contributions clarified when longitudinal models gain power over ANCOVA and adapted these ideas to run-in trials.

The debate between change-score and ANCOVA analyses has a long *history[1–3].[4] provided a formal efficiency comparison for pretest-posttest designs. The closest antecedent of the present work is[5], who derived the relative efficiency of post-treatment, change-score, and ANCOVA analyses when several baseline measurements are available, and[6], who extended the treatment to repeated post-treatment measurements. Their ordering, that ANCOVA on the mean of the available baselines dominates the change score which in turn dominates the raw post-treatment comparison, is the qualitative result that the ordering established in Section 3.1 specializes. What is new here is not the ordering itself but the closed-form relative efficiencies in the run-in random-slope parameterization: the efficiency margins are obtained as explicit functions of the design counts* (J_0, J_1, J_2) *and the signal-to-noise ratio* σ_b/σ*, they cover a post-treatment common close phase absent from the Frison-Pocock setting, and the estimand is a difference of slopes rather than a difference of means. Within the run-in literature,[7] derived GLS power formulas for trials with pre-randomization observations, which the present comparison takes as the efficient benchmark against which the simpler estimators are measured.[8] and[9] compared baseline-as-covariate against baseline-as-response (constrained longitudinal data analysis).[10] developed a semiparametric framework for covariate adjustment.[11] showed that MMRM requires time-by-covariate interactions to guarantee power gains over ANCOVA.[12] established the foundational comparison of pretest-posttest designs.[13] addressed conditioning on baseline in longitudinal RCTs.[14] proposed the two-period approach for run-in trials.*

[3] **The averaged change-score estimator is distinct from and less efficient than the constrained longitudinal data analysis estimator, which is excluded from the present comparison.**

- The cLDA estimator places baseline in the response vector under a common-mean constraint.
- It exploits the full covariance structure without specifying the random-effects distribution, ranking between the averaged estimator and GLS.
- A full comparison including cLDA lies beyond the present scope but constitutes a natural extension.

We note that the averaged change-score estimator studied here is distinct from (and less efficient than) the constrained longitudinal data analysis (cLDA) estimator of[15] and[9], which includes baseline in the response vector with a common-mean constraint. The cLDA estimator uses the full covariance structure without requiring specification of the random effects distribution G*, placing it between the averaged estimator and GLS in the efficiency ordering.[16] extended the cLDA framework to pre- and post-treatment measurements with equal baseline assumptions. A full comparison including cLDA is beyond the scope of this paper but represents a natural extension.*

[4] **The longpower package implements the GLS baseline but omits the simpler estimators and the misspecification analysis developed here.**

- It is the CRAN reference for slope-comparison power calculations under the Liu-Liang, Diggle, and Edland formulations.

- Its routines correspond to the GLS estimator under correct specification, matching the Paper 1 baseline.
- It does not cover ANCOVA, the averaged change-score test, or the efficiency loss from misspecified covariance.

The `longpower` R package[17] provides the CRAN reference implementation of power calculations for the GLS / REML-based slope-comparison estimator, covering the Liu-Liang[18], Diggle[19], and Edland[20] formulations. These routines correspond to the $\hat{\gamma}_{GLS}$ estimator studied here under correct specification and constitute the Paper 1 baseline extended by run-in and common close observations. However, `longpower` does not cover the ANCOVA-on-run-in-mean estimator or the averaged change-score t-test, nor does it characterize the efficiency loss from misspecification of G or R. The comparison developed here therefore complements `longpower` by (i) ordering the three candidate estimators under correct specification and (ii) quantifying the efficiency margin that GLS retains under the misspecifications of G and R studied below.

2 Three estimators

2.1 GLS under the mixed model

[5] **The GLS variance for the treatment effect follows from the relevant diagonal element of the inverse information matrix under the marginal covariance.**

- The per-pair variance is the treatment-effect element of $(X'\Sigma^{-1}X)^{-1}$.
- The marginal covariance decomposes as residual plus random-effects components.
- The inverse is obtained efficiently via the Woodbury identity.

From[21], the per-pair variance is the (γ, γ) element of $(X'\Sigma^{-1}X)^{-1}$ with $\Sigma = R + ZGZ'$[22,23], computed via the Woodbury identity.

For the Frost conventional design ($J_0 = 0$, $J_1 = 2$):

$$\hat{\gamma}_{\text{GLS}} = \frac{1}{2t}[(C_{22} + C_{21}) - (C_{12} + C_{11})] \tag{3}$$

with $\text{Var}(\hat{\gamma}_{\text{GLS}}) = (\sigma^2 + 2\sigma_b^2 t^2)/t^2$.

2.2 Averaged change score (t-test)

The averaged change-score estimator ignores the within-subject correlation structure and treats d_{ik} as independent observations:

$$\text{Var}(\hat{\gamma}_{\text{avg}}) = \frac{2}{m}\,\text{Var}(d_{ik}) \tag{4}$$

The variance $\text{Var}(d_{ik})$ depends on the correlation structure but does not require its estimation. As an estimator of γ, the contrast is divided by the design constant c derived in Section 3.1, and all variances, relative efficiencies, and sample sizes reported in this paper are on that common γ scale.

```r
a <- 1.0
b <- 1.0
tt <- 1.0

# gamma-scale rescaling constant (Section 3.1):
# mean accumulated effect over post-randomization visits
cfac <- function(r, k, f, t_interval = 1) {
  t_interval * (k * (k + 1) / 2 + k * f) / (k + f)
}

v_gls <- var_gamma_matrix(2, 2, 0, a, b, tt)
v_avg <- var_gamma_avg(2, 2, 0, a, b, tt) /
  cfac(2, 2, 0, tt)^2
re <- v_gls / v_avg

cat(sprintf('J_0=2, J_1=2: GLS=%.4f, Avg=%.4f, RE=%.4f\n',
            v_gls, v_avg, re))
```

```
## J_0=2, J_1=2: GLS=2.1569, Avg=8.8889, RE=0.2426
```

2.3 ANCOVA with run-in mean as covariate

The ANCOVA approach uses the averaged run-in rate as a baseline covariate without specifying σ_b^2:

$$\bar{Y}_{i,\text{post}} = \alpha + \gamma g_i + \phi \bar{Y}_{i,\text{pre}} + \varepsilon_i \tag{5}$$

The variance of $\hat{\gamma}$ under this model is

$$\text{Var}(\hat{\gamma}_{\text{ANC}}) = \frac{2}{m}\,\text{Var}(\bar{Y}_{\text{post}})\,(1 - \rho^2_{\text{pre,post}}) \tag{6}$$

[6] **The ANCOVA variance is governed by the run-in to post-randomization correlation, which determines the magnitude of the variance reduction.**

- The factor $(1-\rho^2)$ scales the variance by the squared marginal correlation between the phase means.
- With no run-in observations the covariate vanishes and ANCOVA reduces to the averaged estimator.
- A large correlation, arising when the random-slope variance dominates the residual, yields substantial variance reduction.

where $\rho_{pre,post}$ is the marginal correlation between the run-in and post-randomization means. When $J_0 = 0$ there is no covariate and ANCOVA *reduces to the averaged estimator. When $\rho_{pre,post}$ is large (i.e., when σ_b^2 dominates σ^2), the $(1-\rho^2)$ factor produces substantial variance reduction.*

[7] **ANCOVA occupies a middle ground by estimating the baseline coefficient from the data while still using the predictive run-in observations.**

- Unlike GLS, the coefficient on the run-in mean is estimated empirically rather than derived from assumed variance components.
- Unlike the averaged estimator, it exploits the predictive value of the run-in observations.
- It therefore reduces variance without committing to a full random-effects specification.

The key advantage over GLS is that ϕ is estimated from the data rather than derived from assumed variance components[1,24]. The key advantage over the averaged estimator is that it exploits the predictive value of the run-in observations[2,25].

```r
c220 <- cfac(2, 2, 0, tt)
v_gls <- var_gamma_matrix(2, 2, 0, a, b, tt)
v_anc <- var_gamma_ancova(2, 2, 0, a, b, tt) / c220^2
v_avg <- var_gamma_avg(2, 2, 0, a, b, tt) / c220^2

cat(sprintf('J_0=2, J_1=2 (sigma_b/sigma = 1):\n'))
```

```
## J_0=2, J_1=2 (sigma_b/sigma = 1):
```

```r
cat(sprintf('  GLS:    %.4f\n', v_gls))
```

```
##   GLS:    2.1569
```

```r
cat(sprintf('  ANCOVA: %.4f\n', v_anc))
```

```
##   ANCOVA: 2.9630
```

```r
cat(sprintf('  Avg:    %.4f\n', v_avg))
```

```
##   Avg:    8.8889
```

```r
cat(sprintf('  RE(GLS/ANC): %.3f\n', v_gls / v_anc))
```

```
##   RE(GLS/ANC): 0.728
```

```r
cat(sprintf('  RE(ANC/Avg): %.3f\n', v_anc / v_avg))
```

```
##   RE(ANC/Avg): 0.333
```

2.4 SUR (seemingly unrelated regression)

The SUR approach treats the run-in and treatment periods as separate regression equations linked by correlated errors:

$$Y_1 = X_2 \begin{pmatrix} \beta_{11} \\ \beta_{12} \end{pmatrix} + e_1 \qquad \text{(both groups)} \tag{7}$$

$$Y_2 = X_1 \alpha_2 + e_2 \qquad \text{(baseline)} \tag{8}$$

The deviation-from-mean parameterization gives $\beta_1 = (\delta_1 + \delta_2)/2$ and $\beta_2 = (\delta_2 - \delta_1)/2$, where δ_j are the group-specific rates.

The SUR variance derivation and its comparison to the GLS and averaged estimators are not developed in this paper; the subsection is retained as a specification only and is flagged as future work in the Discussion.

2.5 BLUP as theoretical benchmark

[8] **The three estimators are linear functions of the change scores and admit a common benchmark in the best linear unbiased predictor of the random slopes.**

- Each estimator can be viewed as a linear combination of the observed change scores.
- The BLUP of the subject-specific random slope provides the natural reference predictor.
- It weights the residuals by the inverse marginal covariance using the GLS fixed-effect estimate.

The three estimators above are linear functions of the observed change scores and can be compared to a common benchmark: the best linear unbiased predictor (BLUP) of the individual random slopes. Under the random-slopes model with marginal covariance $V_i = R_i + Z_i G Z_i'$, the BLUP of the subject-specific random slope b_i given the data is

$$\hat{b}_i = G Z_i' V_i^{-1} (y_i - X_i \hat{\beta}) \tag{9}$$

where $\hat{\beta}$ is the GLS estimator of the fixed effects[26,27]*.*

[9] **GLS attains the minimum variance among linear unbiased estimators when the covariance is correct, while the simpler estimators are specific approximations to this benchmark.**

- The group-specific slope estimators inherit the variance of the treatment-effect fixed effect.
- GLS is optimal among linear unbiased estimators under a correctly specified marginal covariance.
- The averaged and ANCOVA estimators are particular linear approximations to this optimal benchmark.

The group-specific slope estimators derived from the fixed effects share the same variance as $(\hat{\beta})_\gamma$, and the GLS estimator $\hat{\gamma}_{GLS}$ minimizes the variance among all linear unbiased estimators of γ when V_i is correctly specified. The averaged and ANCOVA estimators are specific linear approximations to this benchmark:

- **Averaged change-score** replaces V_i^{-1} with an unweighted sum; this is optimal only when $V_i \propto I$, which fails for any non-zero random-slope variance.
- **ANCOVA** collapses the run-in visits into a single scalar covariate (the run-in mean) rather than weighting them by V_i^{-1}; the resulting efficiency loss vanishes as $J_0 \to \infty$ since the run-in mean becomes a sufficient statistic for the subject-specific intercept plus average slope.

[10] **Misspecification erodes but, in the settings studied here, does not eliminate the GLS efficiency advantage over the simpler approximations.**

- GLS propagates errors in the assumed covariance through its inverse weighting.
- The averaged and ANCOVA estimators depend on fewer assumed variance components.
- Section 4 quantifies the resulting erosion; in the misspecifications examined, GLS retains a large margin.

Section 4 examines whether model misspecification erodes the nominal optimality of GLS relative to these approximations, which depend on fewer variance components. In the G and R misspecifications studied there, the GLS variance increases but remains below that of the averaged estimator; whether feasible GLS with estimated components loses its advantage in small samples is not studied here.

3 Relative efficiency under correct specification

3.1 Analytical comparison

The relative efficiency of the averaged estimator to GLS is:

$$\mathrm{RE}_{\mathrm{avg}} = \frac{\mathrm{Var}(\hat{\gamma}_{\mathrm{GLS}})}{\mathrm{Var}(\hat{\gamma}_{\mathrm{avg}}/c)} \leq 1 \tag{10}$$

with equality when the averaged estimator happens to coincide with GLS (which occurs in special cases).

The ordering asserted above can be stated and proved directly. One preliminary is needed. As defined, the averaged and ANCOVA contrasts average the accumulated treatment effect over the post-randomization visits, so their expectation is $c\gamma$ rather than γ, where

$$c = t\,\frac{J_1(J_1+1)/2 + J_1 J_2}{J_1 + J_2} \tag{11}$$

is the mean of the accumulated-effect design column over the post-randomization visits ($c = 1.5t$ for the $(2, 2, 0)$ design; $c = t$ when $J_1 = 1$, $J_2 = 0$). Dividing either contrast by c yields an unbiased estimator of γ with variance scaled by $1/c^2$.

Proposition. Under the random-slopes model with correctly specified Σ, $\mathrm{Var}(\hat{\gamma}_{\mathrm{GLS}}) \le \mathrm{Var}(\hat{\gamma}_{\mathrm{ANC}}/c) \le \mathrm{Var}(\hat{\gamma}_{\mathrm{avg}}/c)$.

Proof. Each rescaled contrast is a linear unbiased estimator of γ, and $\hat{\gamma}_{\mathrm{GLS}}$ is the best linear unbiased estimator of γ when Σ is known (Gauss-Markov), which gives the first inequality. For the second, both contrasts have the form (post-randomization mean) $-\phi\times$ (run-in mean): the population ANCOVA coefficient minimizes the residual variance over ϕ, while the averaged estimator fixes $\phi = 1$, so the ANCOVA contrast variance is no larger; dividing both by c^2 preserves the inequality. (The feasible ANCOVA estimates ϕ and carries the usual finite-sample inflation.) □

The ordering itself is not new.[5] established the same qualitative ranking for repeated-measures designs with several baseline measurements, and[6] extended it to repeated post-treatment measurements; the argument above is the specialization of that result to the run-in random-slope model, where the run-in mean plays the role of their baseline mean. The contribution of the following sections is the closed-form magnitude of the two gaps as functions of (J_0, J_1, J_2) and σ_b/σ, which the Frison-Pocock parameterization does not deliver for a slope estimand or for designs with a common close phase.

Numerically, the ordering holds with zero violations over the 115-cell grid of the tables below. All tables, figures, and simulation results in this paper are on the equal-estimand γ scale: the averaged and ANCOVA contrast variances are divided by c^2 before any comparison. For reference, the raw phase-mean contrasts carry variances larger by the factor c^2 ($c^2 = 2.25$ at the $(2, 2, 0)$ design), so raw-contrast comparisons would overstate the penalty of the simpler estimators.

3.2 When does the gap exceed 5%?

The answer, computed over the full grid of the preceding tables (all on the γ scale), is: in most cells. Of the 115 design-by-ratio cells, 12 have ANCOVA relative efficiency of at least 0.95 (an efficiency loss below 5%), and they form exactly two families. The first (6 cells) comprises single-post-visit run-in designs with no common close ($J_1 = 1$, $J_2 = 0$): the $(1, 1, 0)$ design at every tabulated ratio, plus $(2, 1, 0)$ at $\sigma_b/\sigma = 0.5$; with a single post-randomization visit there is no within-period slope information for GLS to exploit. The second (6 cells) comprises designs without run-in or common close ($J_0 = 0$, $J_2 = 0$, $J_1 \in \{2, 3\}$) at $\sigma_b/\sigma \ge 2$, where ANCOVA reduces to the averaged estimator and the rescaled phase mean is nearly efficient. In run-in designs with $J_1 \ge 2$, the regime that motivates this paper, the loss is substantial: at the representative $(2, 2, 0)$ design the relative efficiency ranges from 0.81 at $\sigma_b/\sigma = 0.5$ down to 0.64 at $\sigma_b/\sigma = 20$, an efficiency loss of 19% to 36%. The averaged estimator reaches 0.95 only in the second family (6 of 115 cells).

Table 1: RE of averaged change-score vs GLS on the γ scale ($\mathrm{Var}_{\mathrm{GLS}}$ / $\mathrm{Var}(\hat{\gamma}_{\mathrm{avg}}/c)$) by σ_b/σ.

(J_0, J_1, J_2)	0.5	1	2	5	20
(1,1,0)	0.667	0.500	0.250	0.056	0.004
(2,1,0)	0.653	0.344	0.117	0.021	0.001
(4,1,0)	0.427	0.149	0.041	0.007	0.000
(0,2,0)	0.818	0.900	0.964	0.994	1.000
(1,2,0)	0.529	0.410	0.204	0.045	0.003
(2,2,0)	0.490	0.243	0.080	0.014	0.001
(4,2,0)	0.276	0.089	0.024	0.004	0.000
(0,3,0)	0.771	0.900	0.969	0.995	1.000
(1,3,0)	0.501	0.394	0.193	0.042	0.003
(2,3,0)	0.433	0.209	0.068	0.012	0.001
(4,3,0)	0.220	0.069	0.018	0.003	0.000
(0,1,1)	0.727	0.533	0.245	0.050	0.003
(1,1,1)	0.452	0.258	0.093	0.017	0.001
(2,1,1)	0.422	0.181	0.055	0.009	0.001
(4,1,1)	0.255	0.080	0.021	0.003	0.000
(0,2,1)	0.705	0.517	0.224	0.044	0.003
(1,2,1)	0.448	0.262	0.094	0.017	0.001
(2,2,1)	0.401	0.172	0.052	0.009	0.001
(4,2,1)	0.221	0.068	0.018	0.003	0.000
(0,3,1)	0.685	0.488	0.202	0.039	0.003
(1,3,1)	0.446	0.258	0.092	0.017	0.001
(2,3,1)	0.380	0.162	0.049	0.008	0.001
(4,3,1)	0.194	0.059	0.016	0.003	0.000

Table 2: RE of ANCOVA vs GLS on the γ scale ($\mathrm{Var}_{\mathrm{GLS}} \,/\, \mathrm{Var}(\hat{\gamma}_{\mathrm{ANC}}/c)$) by σ_b/σ.

(J_0, J_1, J_2)	0.5	1	2	5	20
(1,1,0)	1.000	1.000	1.000	1.000	1.000
(2,1,0)	0.971	0.909	0.860	0.838	0.834
(4,1,0)	0.838	0.747	0.713	0.702	0.700
(0,2,0)	0.818	0.900	0.964	0.994	1.000
(1,2,0)	0.858	0.866	0.853	0.838	0.834
(2,2,0)	0.808	0.728	0.671	0.648	0.643
(4,2,0)	0.637	0.544	0.512	0.502	0.500
(0,3,0)	0.771	0.900	0.969	0.995	1.000
(1,3,0)	0.808	0.815	0.785	0.758	0.751
(2,3,0)	0.732	0.642	0.582	0.558	0.553
(4,3,0)	0.538	0.450	0.420	0.411	0.409
(0,1,1)	0.727	0.533	0.245	0.050	0.003
(1,1,1)	0.733	0.545	0.390	0.319	0.304
(2,1,1)	0.697	0.542	0.460	0.430	0.425
(4,1,1)	0.587	0.490	0.456	0.446	0.445
(0,2,1)	0.705	0.517	0.224	0.044	0.003
(1,2,1)	0.723	0.541	0.385	0.313	0.299
(2,2,1)	0.678	0.527	0.447	0.418	0.412
(4,2,1)	0.539	0.444	0.413	0.403	0.401
(0,3,1)	0.685	0.488	0.202	0.039	0.003
(1,3,1)	0.699	0.516	0.362	0.293	0.279
(2,3,1)	0.642	0.494	0.416	0.388	0.382
(4,3,1)	0.486	0.397	0.367	0.359	0.357

4 Robustness to misspecification

4.1 Misspecified G

[11] **Misestimating the random-slope variance forfeits the optimality of GLS while leaving the averaged estimator unchanged.**

- An incorrect random-slope variance renders the assumed-covariance GLS no longer minimum-variance.
- The error enters through the inverse-covariance weighting of GLS.
- The averaged change-score estimator's definition does not use this variance component, so its variance does not depend on the assumed value.
- In the sweep below, however, the misspecified GLS variance peaks at 5.30 while the averaged variance on the γ scale is 8.89: the GLS margin is eroded, not eliminated.

When σ_b^2 is misestimated (e.g., too large or too small), the GLS analysis that assumes it is no longer optimal. Its sampling variance is then the sandwich form $(X'\Sigma_a^{-1}X)^{-1}X'\Sigma_a^{-1}\Sigma\Sigma_a^{-1}X(X'\Sigma_a^{-1}X)^{-1}$, with Σ_a the assumed and Σ the true covariance, which is the model-robust variance of[28] evaluated at a known working covariance rather than an estimated one. The averaged change-score estimator's definition does not depend on the assumed value, so its variance is constant across the sweep. Note the scale of the comparison: over assumed $\sigma_b^2 \in [0.01, 5]$ with true $\sigma_b^2 = 1$, the misspecified GLS sandwich variance never exceeds 5.30, still below the averaged-estimator variance of 8.89 on the γ scale. The studied misspecification erodes but does not overturn the GLS advantage.

4.2 Misspecified R

[12] **Misspecifying the residual correlation biases the GLS variance machinery but again leaves the averaged estimator intact.**

- Assuming compound symmetry when the truth is AR(1) is a representative misspecification.
- Both the GLS variance formula and the Woodbury correction become biased.
- The averaged estimator's definition does not depend on the assumed correlation, although its γ-scale variance still changes with the true R (8.4 to 9.1 over the plotted range).
- The misspecified GLS sandwich variance stays four- to elevenfold below the averaged variance across the sweep.

When the residual correlation structure is wrong (e.g., compound symmetry assumed but AR(1) is true), both the GLS variance formula and the Woodbury correction are biased. The averaged estimator's definition does not depend on the assumed model, though its γ-scale variance does change with the true

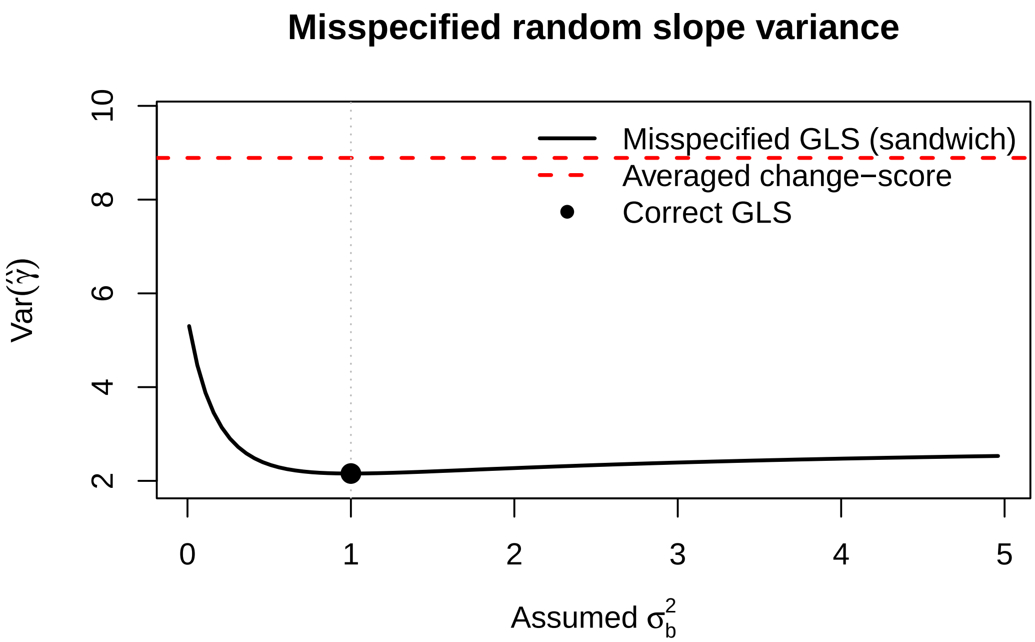

Figure 1: (#fig:misspec_G)Effect of misspecifying σ_b^2 on GLS variance. True $\sigma_b^2 = 1$. The GLS estimator uses an assumed value on the x-axis. The averaged estimator's definition does not use σ_b^2, so its γ-scale variance (horizontal line) does not depend on the assumed value. The misspecified GLS variance remains below the averaged variance throughout.

R (8.4 to 9.1 across $\rho \in [0, 0.9]$ in the figure). The GLS sandwich variance ranges from 0.74 to 2.16 over the same sweep, so the studied R misspecification leaves GLS four to eleven times more efficient than the averaged estimator throughout.

5 Simulation study

5.1 Design

[13] **A factorial Monte Carlo design complements the analytical results by varying sample size, design, signal-to-noise, and analysis method under a compound-symmetric truth.**

- Sample size and the run-in design configuration are crossed.
- The signal-to-noise ratio is varied; the true correlation is compound symmetric throughout.
- Three analysis methods are compared: correctly specified GLS, misspecified GLS, and the averaged change-score test.

To complement the analytical results, we simulate trials under the following factorial design:

- $m \in \{50, 100\}$ per group
- $(J_0, J_1, J_2) \in \{(0, 2, 0), (2, 2, 0), (2, 2, 2)\}$
- $\sigma_b/\sigma \in \{1, 5\}$

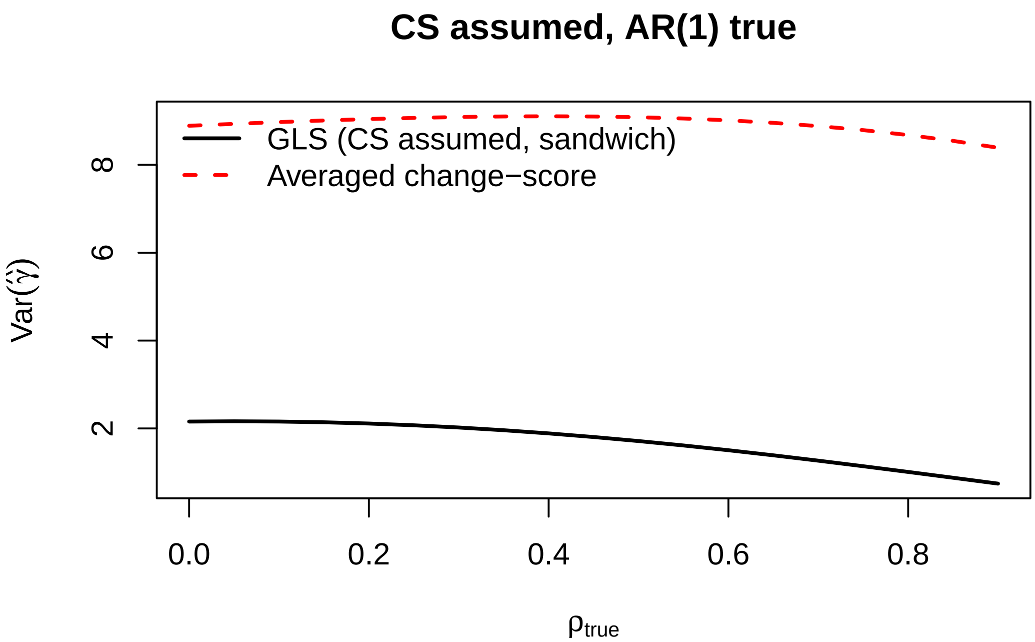

Figure 2: (#fig:misspec_R)Effect of analyzing under compound symmetry when the true correlation is AR(1). Sandwich variance of GLS (solid) vs averaged change-score on the γ scale (dashed) as a function of ρ. The averaged estimator's definition does not use the assumed correlation, but its variance varies with the true R; GLS remains at least fourfold more efficient at every ρ.

- True correlation: compound symmetry (an AR(1) truth is examined analytically in Section 4 but is not simulated)
- Analysis: GLS (correctly specified), GLS (misspecified), averaged change-score

[14] **Each configuration is replicated 1000 times and evaluated on empirical variance and power; coverage is not assessed.**

- One thousand trials are simulated per configuration.
- Empirical standard deviation and empirical power are recorded for each estimator.
- Coverage of nominal 95% confidence intervals, empirical size at $\Delta = 0$, larger sample sizes, higher ratios, an AR(1) truth arm, and 5000 replicates remain future work.

For each configuration, 1000 trials are simulated. The outcomes are the empirical standard deviation of $\hat{\gamma}$ and the empirical power. Coverage of nominal 95% confidence intervals and empirical size at $\Delta = 0$ are not assessed; extending the factorial to $m \in \{30, 200\}$, $\sigma_b/\sigma = 20$, an AR(1) truth arm, and 5000 replicates remains future work. Test statistics use standard normal critical values rather than t; the distinction is immaterial at $m \geq 50$ per group.

5.2 Results

[15] **The simulation confirms the analytic ordering and quan-**

Table 3: Monte Carlo comparison of GLS (correctly specified), GLS (misspecified σ_b^2, assuming $\sigma_b^2 = \sigma^2$), and the averaged change-score estimator under compound symmetry (1000 replicates, $\Delta = 0.5$, two-sided $\alpha = 0.05$). Columns report the empirical standard deviation of $\hat{\gamma}$ and the empirical power. The averaged estimator is rescaled to the γ scale (contrast divided by c).

Design	σ_b/σ	m	SD GLS	SD mis	SD avg	Pow GLS	Pow mis	Pow avg
(0,2,0)	1	50	0.238	0.238	0.250	0.537	0.537	0.495
(2,2,0)	1	50	0.210	0.210	0.417	0.677	0.677	0.206
(2,2,2)	1	50	0.151	0.151	0.458	0.902	0.902	0.192
(0,2,0)	1	100	0.169	0.169	0.178	0.823	0.823	0.769
(2,2,0)	1	100	0.146	0.146	0.308	0.926	0.926	0.393
(2,2,2)	1	100	0.108	0.108	0.334	0.997	0.997	0.345
(0,2,0)	5	50	1.012	1.012	1.017	0.083	0.653	0.083
(2,2,0)	5	50	0.241	0.528	2.043	0.555	0.600	0.067
(2,2,2)	5	50	0.164	0.284	2.297	0.873	0.765	0.054
(0,2,0)	5	100	0.726	0.726	0.726	0.108	0.733	0.109
(2,2,0)	5	100	0.162	0.360	1.387	0.857	0.748	0.067
(2,2,2)	5	100	0.106	0.189	1.573	0.997	0.936	0.060

tifies the cost of misspecifying the random-slope variance.

- The empirical standard deviation of the GLS estimator is the smallest of the three under correct specification.
- On the γ scale the averaged estimator is nearly as efficient as GLS in the no-run-in design, but carries two to three times the GLS standard deviation in the run-in designs at $\sigma_b/\sigma = 1$, growing to eight to fifteen times at $\sigma_b/\sigma = 5$.
- Misspecifying the random-slope variance at most roughly doubles the GLS standard deviation, which remains well below that of the averaged estimator in every run-in cell.

We simulate the factorial design described above under compound symmetry, comparing the correctly specified GLS estimator, a GLS estimator that misstates the random-slope variance (it assumes $\sigma_b^2 = \sigma^2$ regardless of the truth), and the averaged change-score estimator, the latter rescaled to the γ scale by dividing the contrast by c. For each cell we report the empirical standard deviation of $\hat{\gamma}$ and the empirical power from 1000 replicates; the Monte Carlo standard error of a tabulated power estimate is at most $\sqrt{0.25/1000} \approx 0.016$.

One feature of the table requires comment. In the $(0, 2, 0)$ rows the correctly specified and misspecified GLS analyses report identical empirical standard deviations but different powers. With $J_0 = 0$ the design has two observations and two fixed effects, so the GLS point estimator does not depend on the assumed covariance and the two analyses coincide as estimators; the apparent power advantage of the misspecified analysis reflects only an anticonservative model-based standard error, not additional information. More generally, because empirical size at $\Delta = 0$ and interval coverage are not tabulated, power comparisons across variance models in

Table 4: Sample size per group: GLS, ANCOVA, and averaged estimators (MIRIAD parameters, $\Delta = 0.25$, 90% power). All variances are on the γ scale (ANCOVA and averaged contrasts divided by c).

(J_0, J_1, J_2)	N GLS	N ANC	N Avg
(4,3,1)	36	95	1630
(4,3,0)	43	100	1676
(4,2,1)	60	144	2413
(2,3,1)	69	162	1053
(4,2,0)	78	151	2368
(2,3,0)	93	157	1028
(1,3,1)	99	255	821
(2,2,1)	104	226	1479
(0,3,1)	112	433	433
(1,2,1)	138	335	1116
(2,2,0)	145	213	1361
(0,2,1)	150	527	527
(4,1,1)	156	339	5328
(1,3,0)	185	233	775
(2,1,1)	226	478	3063

this table are not interpretable as efficiency comparisons; adding size and coverage columns is remaining work.

6 Numerical examples

6.1 MIRIAD parameters

6.2 When to use which estimator

A decision rule based on m, σ_b/σ, and confidence in the correlation structure requires the feasible-GLS simulation described in the Discussion and is deferred to a revision. The results above support only the narrower statement that, with variance components taken as known, GLS dominates and the penalty of the simpler estimators is largest in run-in designs with $J_1 \geq 2$ and high σ_b/σ.

7 Discussion

[16] **GLS is the efficient choice under a correctly specified, adequately estimated model, which in practice means moderate-to-large samples.**

- GLS attains the minimum variance among linear unbiased estimators when the covariance is known.
- Its advantage is realized only when the variance components are estimated accurately, which requires sufficient sample size and follow-up.

- In small samples the instability of the estimated covariance can erase the theoretical gain.

The estimator comparison yields a graded recommendation rather than a single winner. Under correct specification GLS is efficient and dominates both ANCOVA and the averaged change-score estimator, as the ordering $\mathrm{Var}_{GLS} \leq \mathrm{Var}_{ANC} \leq \mathrm{Var}_{avg}$ *confirms. This dominance is a large-sample statement: it presumes the covariance components are known or well estimated. With moderate-to-large samples and a credible covariance model, GLS is the appropriate default.*

[17] **The misspecifications studied here erode the GLS advantage without overturning it; the open robustness question concerns feasible GLS with estimated components.**

- GLS weights observations by the assumed inverse covariance, so errors in G or R propagate into its sampling variance.
- In the sweeps of Section 4 the misspecified GLS sandwich variance peaks at 5.30 against a γ-scale averaged-estimator variance of 8.89, and in the simulation the misspecified GLS standard deviation remains well below the averaged standard deviation in every run-in cell.
- Whether feasible GLS, with variance components estimated from small samples, retains this margin is not studied here.

The misspecification analysis qualifies the case for GLS less than might be expected. Because GLS weights the data by an assumed inverse covariance, errors in the assumed G or R feed into its sampling variance, and the sandwich-variance calculations quantify the inflation. In the settings studied, however, the inflation never closes the gap: the misspecified GLS variance peaks at 5.30 in the G sweep and at 2.16 in the R sweep, against γ-scale averaged-estimator variances of 8.4 to 9.1 at the same design. The averaged change-score estimator avoids the variance components entirely, and ANCOVA estimates the run-in coefficient from the data, but under the misspecifications examined neither robustness property is enough to overcome their efficiency deficit. These calculations take the variance components of the GLS analysis as fixed inputs; the open and practically important question is whether feasible GLS, with components estimated by REML from small samples, keeps its advantage. That question is not answered here.

[18] **The efficiency gap between GLS and the simpler estimators narrows as the design grows, so the choice matters most in small, information-poor designs.**

- With many run-in and treatment observations the phase means become near-sufficient for the random slope.
- The averaged and ANCOVA estimators then recover most of the information GLS extracts, and the relative efficiency approaches one.
- The estimator choice is therefore most consequential when observations are few and the slope-to-noise ratio is high.

The practical importance of the estimator choice depends on the design. As the numbers of run-in and treatment observations grow, the phase means approach sufficiency for the subject-specific slope, the ANCOVA efficiency loss vanishes as $J_0 \to \infty$, and the relative efficiencies of all three estimators converge toward one. The gap is largest, and the choice of estimator therefore most consequential, in small designs with few observations per phase and a high slope-to-noise ratio, exactly the regime in which the estimated covariance is also least stable. Whether that instability erodes the feasible-GLS advantage enough to change the ranking in small studies is the question the planned feasible-GLS simulation must answer; the known-components results here cannot.

[19] **Two extensions, the seemingly unrelated regression estimator and the constrained longitudinal data analysis estimator, would complete the efficiency picture.**

- The SUR formulation offers a middle ground between the averaged estimator and full GLS but adds implementation complexity.
- The cLDA estimator exploits the full covariance without specifying the random-effects distribution and is expected to rank between ANCOVA and GLS.
- A full comparison including both, and a feasible-GLS simulation under estimated components, is the natural continuation of this work.

Two estimators not fully developed here would round out the comparison. The seemingly unrelated regression formulation of Section 2.4 treats the run-in and treatment periods as linked equations and offers a compromise between the averaged estimator and full GLS, at the cost of additional implementation complexity; its variance derivation remains to be completed. The constrained longitudinal data analysis estimator uses the full covariance without committing to a random-effects distribution and should fall between ANCOVA and GLS in efficiency. A comprehensive treatment would compare all of these under a feasible-GLS simulation in which the variance components are estimated rather than assumed, which is the setting in which the robustness distinctions drawn here become decisive.

8 Conflict of interest

The author declares no conflicts of interest.

9 Funding

This work received no specific funding.

10 Data availability statement

[20] **The package implementing all three estimators, the simulation drivers, and the shared bibliography are openly available in the project repository.**

- The estimator implementations reside in the package source files.
- The simulation drivers and shared compendium bibliography are included.
- Specific paths appear in the Reproducibility section below.

The `runinpower` R package implementing the GLS, ANCOVA, and averaged-change-score estimators (`ancova.R`, `averaged.R`), the simulation drivers, and the shared bibliography across the four-paper compendium are openly available in the project repository. Specific paths are listed in the Reproducibility section below.

11 Reproducibility

[21] **The manuscript was rendered with R 4.4.x using the package source and a small set of standard analysis and build packages.**

- The in-package source supplies the estimator implementations.
- The GLS comparator in the simulation is implemented directly in the manuscript source (the analyze functions of the simulation chunk); tinytest is the testing harness.
- The manuscript is built with rmarkdown and bookdown.

This manuscript was rendered with R 4.4.x. Principal packages used are the in-package `runinpower` source, `tinytest` (testing harness), and `rmarkdown` plus `bookdown` for the manuscript build. The GLS comparator in the simulation is implemented directly in the manuscript source (`analyze_gls` in the simulation chunk); no external mixed model fitter is used.

[22] **The simulation sets a single fixed seed before the replicate loop, which makes the run exactly reproducible.**

- A single seed is set once at the entry point of the Monte Carlo comparison.
- No per-replicate reseeding is performed; the replicate stream follows from the single seed.
- Re-running the chunk therefore reproduces the tabulated results exactly.

Random-number generator. *The Monte Carlo simulation comparing the three estimators sets `set.seed(20260418)` once, immediately before the replicate loop. No per-replicate reseeding is performed, so the full replicate stream, and hence the tabulated results, follow deterministically from that single seed.*

[23] **All code, drivers, companion manuscripts, and the shared bibliography are located at documented paths within the**

compendium.

- The package source and simulation drivers reside in their respective directories.
- The three companion manuscripts are stored alongside this paper in the compendium.
- The manuscript source and shared bibliography are co-located in the paper directory.

Data and code availability. *The R package source is at `R/`. Simulation drivers are at `analysis/scripts/`. The companion manuscripts (Papers 1, 2, 3) are at `analysis/report/{01,02,03}-*/`. The manuscript source and shared bibliography (`references.bib`, a symlink to `../../references.bib`) are at `analysis/report/04-gls-vs-ttest/`.*

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